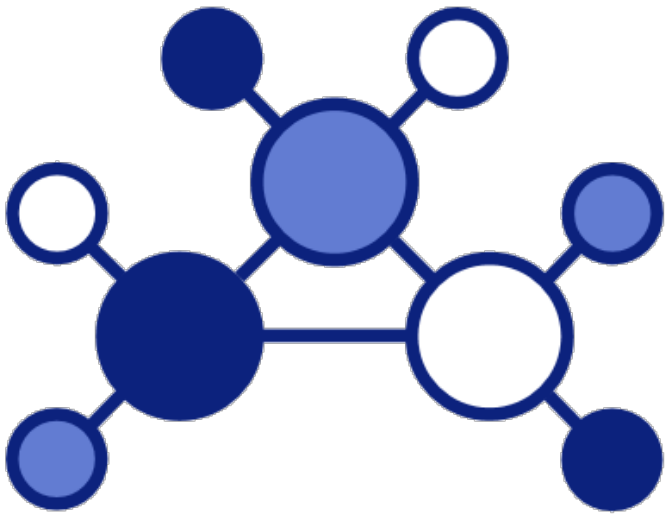

Federated Mission Networking

FMN Spiral 5
Service Instructions for Domain Naming

Disclaimer

This document is part of the Federated Mission Networking (FMN) Spiral 5 Specification. In the FMN management structure it is the responsibility of the Capability Planning Working Group (CPWG) to develop the FMN Spiral Specifications, in close cooperation with the Operational Coordination Working Group (OCWG) and the Multinational CIS Security Management Authority Working Group (MCSMAWG), in order to accomplish the scope from the FMN Spiral Specification Roadmap and support Affiliates' long term capabilities planning.

The FMN Spiral 5 Specification delivers a series of procedural and service instructions to describe what actions FMN Affiliates should undertake to establish their interface with any federation of mission participants in term of People, Processes and Technology. These instructions are completed with documents that provide an overview of the interoperability architecture, requirements, standard profiles, terms and definitions and interface profiles.

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Table of Contents

1 Introduction **5**
1.1 Aim . . . 5
1.2 Scope . . . 5
2 Changes **6**
3 References **8**
4 Context **9**
4.1 Name Space Hierarchy . . . 9
4.2 Content of the Zones . . . 10
4.3 Management of Zones . . . 10
4.4 Service Components . . . 10
4.5 Information Domain Federation . . . 11
4.6 Protected Core Federation . . . 11
5 Interoperability Architecture **13**
5.1 Interactions . . . 13
5.1.1 DNS Query . . . 14
5.1.2 DNS Root . . . 19
5.1.3 DNS Root Zone Transfer . . . 21
5.1.4 Anycast DNS Advertising . . . 24
5.2 Architecture Items . . . 27
5.2.1 Services . . . 27
5.2.2 Data Objects . . . 27
5.3 Profiles . . . 28
5.3.1 Generic Domain Naming Profile . . . 28
5.3.2 IPv6 Domain Naming Profile . . . 28
5.3.3 Anycast DNS Profile . . . 28
5.3.4 Zone Transfer Profile . . . 29
5.3.5 Secure Domain Naming Profile . . . 29
6 Procedures **31**
6.1 Planning Procedures . . . 31
6.1.1 Federated Domain Name Services Framework Process . . . 31
6.1.2 Federated Domain Name Services Design Process . . . 31
6.1.3 Federated Domain Name Services Setup Process . . . 31
6.2 Joining Procedures . . . 32
6.2.1 Federated Domain Name Services Joining Process . . . 32
6.3 Membership Procedures . . . 32
6.3.1 Federated Domain Name Services Membership Process . . . 32
6.3.2 Federated Domain Name Services Change Management Process . . . 32
6.4 Transfer of SMA Procedures . . . 33
6.4.1 Federated Domain Name Services SMA Transfer Process . . . 33
6.5 Exit Procedures . . . 33
6.5.1 Federated Domain Name Services Exiting Process . . . 33

Annex A. Definitions **34**

Annex B. Requirements **35**

B.1 Federation-level Requirements . . . 35

B.2 Enterprise-level Requirements . . . 36

Annex C. Dependencies **37**

C.1 Prerequisites . . . 37

C.2 Usage . . . 39

C.2.1 Supported Services . . . 39

Annex D. Roles and Responsibilities **42**

Annex E. Configuration Parameters **44**

1 Introduction

1.1 Aim

The Service Instructions for Domain Naming provide guidance as to the procedures, supporting services, infrastructure and data attributes required to implement services for the Domain Name System (DNS) in federated Mission Networks.

As such, this document contributes to the establishment of capabilities in support of Federated Mission Networking (FMN) as an affordable, effective and efficient means to enable sharing of information in a coalition environment.

1.2 Scope

These service instructions focus on the description of how the Domain Name System (DNS) infrastructure for all conceptual deployments should be designed (including but not limited to Information Domains and a Protected Core Network Domain). The instruction applies to all MN Providers in each independent domain.

The instructions contain the information to setup the Domain Name Service in a federated network. The main objectives of this solution for the Domain Name Services are:

- To guarantee a proper level of authority and independence to FMN mission network providers;
- To not require reconfiguration when circumstances change, and to be flexible and robust at the same time;
- To be easy to implement and maintain;
- To comply with the Protected Core Networking (PCN) concept whenever applicable.

These service instructions also describe the context how Domain Naming is to be applied to information security domains and the Protected Core.

These instructions do not describe:

- The DNS standard and functionality in great detail;
- Any specific DNS implementation; or
- Aspects related to implementation of name servers, i.e.:
  - The implementation of server redundancy (master/slave server); or
  - The implementation of domain name caching architecture and procedures.

2 Changes

The principle on which capability planning is based in the context of Federated Mission Networking is incremental development. A time boxed planning is used to produce FMN Spiral Specifications that inspire a gradual but steady maturation of the FMN capabilities in two ways: either by the introduction of new Capability Enhancements (CPEs) on the basis of Spiral Objectives (SPOs), or through an continuous effort to improve existing capabilities in the FMN Framework. In other words, each FMN Spiral Specification invokes changes to increase federated interoperability.

This chapter provides an overview of the scale of change compared to the previous spiral specification with the following details:

- Summary -- a text segment with a summary of the changes in these service instructions compared to the previous spiral.
- Capability Enhancements -- a table listing all CPEs that are fulfilled with these service instructions, each with its own summary.
- Backwards Compatibility -- a text segment describing the compatibility of the implemented changes in this specification in relation to the previous spiral.
- Resolved Issues -- a bullet list of the issues that have been raised on these service instructions and that have been resolved within, displayed with issue number and title. The problem statements, discussions and ultimate resolution of these issues can be consulted in the Issue Tracker.

Summary

- Mandated the usage of a signed root zone.
- Signing algorithms for DNSSEC have been constrained to a suite of algorithms.
- Clarified applicability of anycast routing (mandatory for root zone).
- Removed mentions of pre-configured root zone.
- Mandated provision of two independent name servers for each delegated zone.
- Clarified usage of zone change notifications (NOTIFY) to speed up zone transfers.
- Mandated usage of Secret Key Transaction Authentication for DNS (TSIG) for zone transfers.
- Re-designed management process.

Capability Enhancements

Capability Enhancement	Summary
Mandate the usage of a signed root zone in Domain Naming to enable DNSSEC for Mission Network Participants (CPE-469)	Mandated the usage of a signed root zone. Signing algorithms for DNSSEC have been constrained to a suite of algorithms.

Backwards Compatibility

To be fully compatible with Spiral 5, the SMA/Root zone operator is required to be able to sign the root zone. Root Servers are required to be able to respond with DNSSEC signed responses.

Technically, DNSSEC support is assumed to be available for all non-historical implementations of authoritative name servers. Missing DNSSEC support in the root zone - be it by a missing technical capability or by SMA decision - will lead to a working system, but may degrade the service quality and security of mission network participants that are willing to sign validate DNS records.

Usage of DNSSEC outside of the root zone is enabled by the newly introduced changes of FMN Spiral 5, but "normal" mission network participants (i.e. those that are not root zone providers) are still free to chose whether they want to use the new capabiliy or not.

Resolved Issues

- **6808**: Preconfigured Root with well-known name-servers *(major)*
- **6815**: Add TSIG for DNS zone transfers *(normal)*
- **6836**: Add IPv6 support in Domain Naming for PCN *(enhancement)*
- **6837**: Mandate the usage of a signed root zone in Domain Naming to enable DNSSEC for Mission Network Participants *(enhancement)*
- **7049**: SREQ-613 definition of two independent servers *(enhancement)*

- **7050**: SREQ-614: Classfull IP terminology in rDNS requirement *(enhancement)*
- **7720**: Improved Management Processes for Domain Naming *(enhancement)*
- **7968**: CPWG Support Team Coherence and Quality Assurance Review: SI for Domain Naming *(enhancement)*
- **8201**: Remove standard RFC 1135 from DOB-17 DNS Object *(normal)*
- **8722**: DNSSEC cryptographic algorithms *(enhancement)*
- **8754**: Wording change for SREQ-1159 RMK-400 *(normal)*
- **8839**: Typo on Page 14 *(enhancement)*
- **9010**: ROU review of DNS SI *(enhancement)*
- **9031**: NCS Review of S5S DNS SI *(normal)*

3 References

This chapter lists the source documentation that has been used as a reference in the development of these service instructions. Typically, the list includes the identifiers and titles of related documents and files, as well as standard specifications.

The metadata of all these references (including title, description, identifying code, publication date, classification and version) is available on the spiral wiki, often accompanied by the original file and/or a link to the source of that reference.

The list below offers the references specifically for these service instructions. In order to put the references in the right context, they are distributed over three distinct categories based upon their use and origin in the specification.

- Normative Standards -- all standards that are specifically defined in Profiles.
- Non-normative Standards -- all references that are defined in Profiles.
- Non-normative References -- all references that are defined in the instructions itself, in data objects and in term.

The complete set of references for the spiral specification is listed in an annex to the main document.

Normative Standards

- RFC 1034 - Domain names - concepts and facilities
- RFC 1035 - Domain names - implementation and specification
- RFC 1123 - Requirements for Internet Hosts - Application and Support
- RFC 2181 - Clarifications to the DNS Specification
- RFC 2782 - A DNS RR for specifying the location of services (DNS SRV)
- RFC 3258 - Distributing Authoritative Name Servers via Shared Unicast Addresses
- RFC 3596 - DNS Extensions to Support IP Version 6
- RFC 4033 - DNS Security Introduction and Requirements
- RFC 4034 - Resource Records for the DNS Security Extensions
- RFC 4035 - Protocol Modifications for the DNS Security Extensions
- RFC 4509 - Use of SHA-256 in DNSSEC Delegation Signer (DS) Resource Records (RRs)
- RFC 4786 - Operation of Anycast Services
- RFC 5155 - DNS Security (DNSSEC) Hashed Authenticated Denial of Existence
- RFC 5702 - Use of SHA-2 Algorithms with RSA in DNSKEY and RRSIG Resource Records for DNSSEC
- RFC 5936 - DNS Zone Transfer Protocol (AXFR)
- RFC 6382 - Unique Origin Autonomous System Numbers (ASNs) per Node for Globally Anycasted Services
- RFC 6724 - Default Address Selection for Internet Protocol Version 6 (IPv6)
- RFC 6891 - Extension Mechanisms for DNS (EDNS(0))
- RFC 7766 - DNS Transport over TCP - Implementation Requirements
- RFC 8945 - Secret Key Transaction Authentication for DNS (TSIG)

Non-normative Standards

- RFC 7094 - Architectural Considerations of IP Anycast

Non-normative References

- RFC 4271 - A Border Gateway Protocol 4 (BGP-4)
- RFC 4949 - Internet Security Glossary, Version 2

4 Context

The service instructions articulate how specific federated military capabilities in a federation of mission networks shall be supported by the design, implementation and deployment of technical components. This chapter describes the technical services, data and their attributes, interoperability standards, configuration parameters and management processes required to provide such capabilities. The business architecture is described further in depending procedural instructions.

The Domain Name Service, based on the Domain Name System (DNS), is a hierarchical distributed name system for IP enabled devices, services and resources. DNS is mostly used to translate easily memorized domain names into numerical IP addresses needed for the purpose of locating computer services and devices on the network. Operation of DNS infrastructure requires proper administration of overall name space as well as existence of authoritative name servers for each zone in the name space. While standard DNS is based on hierarchy or authoritative name servers that have permanent connectivity, a federation needs to support name resolution in isolated islands of networks that merge and disjoin.

The Domain Name System is an essential pre-requisite for some services like email services and web services.

The federated Domain Name Service is based on the Domain Name System, as defined by the Internet Engineering Task Force (IETF). The capability to build ad-hoc federations is achieved having a system of static (across missions), well-known name-spaces in combination with root servers at well-known anycast addresses.

4.1 Name Space Hierarchy

Domain name space is hierarchical structure of name space allocations for individual authoritative nodes that maintain their own records in the name space hierarchy, and are responsible for delegations to the one step lower nodes in the name space hierarchy.

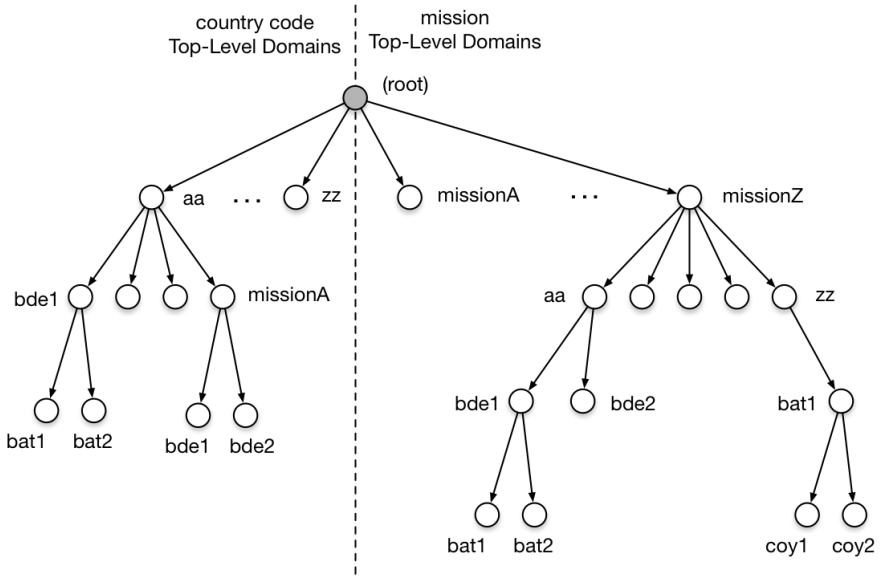

Figure 1. Name Space Hierarchy

Domain hierarchy starts with "Null" or ".". This root of the name space hierarchy contains authoritative records of root name servers as well as zone information, which glues together the top-level name spaces and respective authoritative name servers.

The Domain Name Service can be based on a combination of

- Country code top-level domains in form of .nation.; and
- Mission top-level domains in form of .mission.

In case of country code top-level domains the allocated country code is used as the top-level domain for the participant.

When using mission dependent (two-tier) model, federation participants (nations/organizations) obtain a directly delegated sub-domain of the mission TLD such as “CCC.mission.”

In case of mission TLD federation participant may continue to make use of their national TLDs for technical reasons (e.g. publication of CRL for the Root CA).

4.2 Content of the Zones

Each zone contains information about the location of the zone in the name space hierarchy, validity of zone records in time, authoritative owner (name server) of the zone as well as actual zone records. These records can be:

- A-records defining name to the IPv4 address translation;
- PTR-records defining IP address to name translation;
- CNAME-records defining alias names for name service records;
- SOA-record defining the start of the zone authority;
- NS-records defining the authoritative name server for the zone;
- RRSIG-records defining a cryptographic proof of authenticity of a resource record;
- DS-records defining a delegation of the signer of a subordinate zone;
- MX-records defining the mail server name for the zone; and/or
- SRV-records defining the service dependent server and protocol/port information for the zone.

Each zone contains at least a SOA-record, NS-record and A-record matching to the NS-record. Each zone must also have reverse zone that contains PTR-records matching to the reverse direction of A-records. Reverse zone are glued in to the name space hierarchy with special "in-addr.arpa" domain that represents the dotted decimal notation of IP-address space that is being mapped against particular name space zone.

4.3 Management of Zones

There are two modes of operation for the population of the zone file:

- A statically configured zone; and
- Zone updates via zone transfers.

Typically, an implementation will use a statically configured zone for the autonomous and ad-hoc capability. While a mission deployment may become more permanent, zone transfers allow possibility to automatically distribute changes in the zones.

The root zone is populated with resource records providing delegation of top-level domains. These records are drawn from mission planning documents and documents of well-known addresses of authoritative country code name servers.

4.4 Service Components

The Domain Name Service has three major components:

- Zone data
  - The domain name spaces and resource records, which are specifications for a tree structured name space and data associated with the names. Conceptually, each node and leaf of the domain name space tree names a set of information, and query operations are attempts to extract specific types of information from a particular set.
- Name servers
  - Name server is the DNS entity responding to resolver queries. It holds information about the domain tree's structure and resource records. A name server may cache information about any part of the domain tree, but in general a particular name server has complete information about a subset of the domain space, and pointers to other name servers that can be used to lead to information from any part of the domain tree.
  - An authoritative name server is responsible for set data (zone resource records) associated with one or more zones in the DNS tree structure. An authoritative name server must be present within every DNS Provider and responding to queries concerning (but not limited to) local delegated zone.
- Resolvers
  - A stub resolver is part of an Operating System. DNS requests are received via an API and resolved in recursive mode. In recursive mode all requests are sent to the same set of configured DNS servers. Queries have the “recursive” flag set.
  - Iterative mode is the process of resolving a resource by iteratively querying authoritative name servers, beginning with the DNS root servers, until the requested resources are found.
  - The resolving process described within this document refers only to iterative resolvers - who use iterative mode to resolve recursive queries.

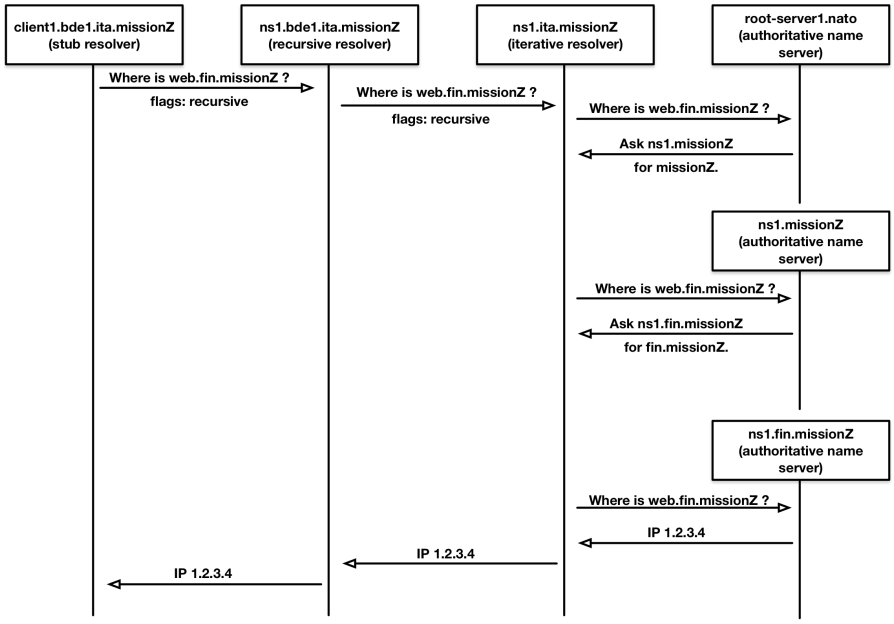

Figure 2. Service Components

4.5 Information Domain Federation

Service Providers in federated Mission Networks have to maintain a separate domain name system in each independent network segment.

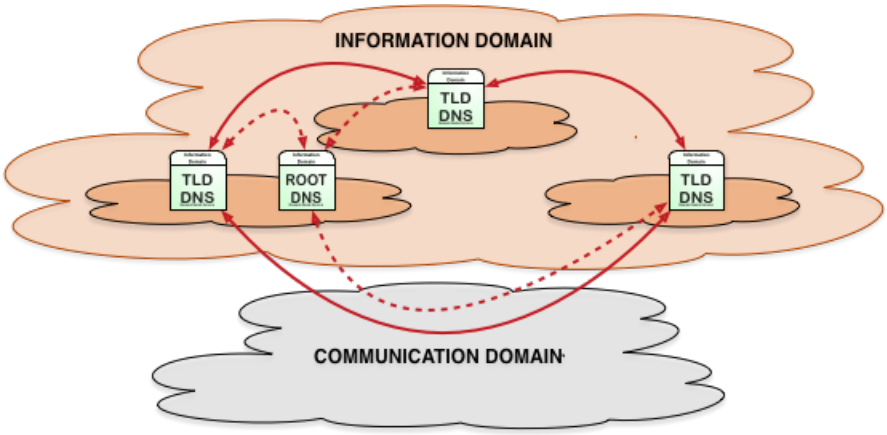

Figure 3. Information Domain Federation

For securing the availability of root zone on all areas of the Mission Network, the design architecture of the Domain Name Service should be such that root zone is anycasted. This is especially important when Mission Network has wide geographical footprint and/or there is low redundancy on interconnecting transport infrastructure. Separate DNS hierarchy is required for each INFOSEC domain Mission Network has.

4.6 Protected Core Federation

Service Providers (PCSs) and Consumers (PCSCs) in federated Mission Networks have to maintain a separate domain name system in the Protected Core.

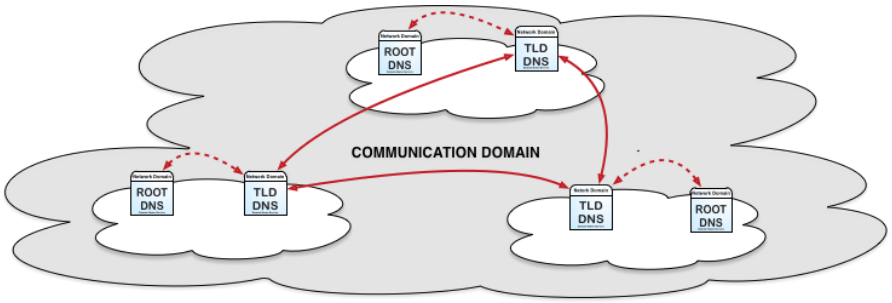

Figure 4. Protected Core Federation

Protected Core name servers must have either a anycasted root, direct forwarders to all other zones or have separate root name server in each Protected Core Segment and Protected Core Service Consumer. Domain Name Server is accessible via the management network of the Protected Core Service Consumer and/or Protected Core Segment as a part of the Network Management and Cyber Defense (NMCD) element.

5 Interoperability Architecture

This chapter provides the interoperability architecture at the basis of these service instructions. It describes the associated Technical Interactions (TINs), and furthermore, architecture items and standards profiles.

- Interactions -- an ArchiMate view displaying all Technology Interactions (TINs), followed by a specified listing per TIN. These contain textual description of the interaction, including possible conditionalities and an ArchiMate view of the complete interaction. Further, it shows:
  - Profiles (PRFs), including their obligation status (Mandatory, Conditional or Emerging) and remarks.
  - Specification Requirements (SREQs), per specific endpoint and within certain dimensions.
  - Configuration Parameters (CFGs), with their identifier (ID), title and description.
  - Prerequisites, in case these are defined on the basis of Dependencies (IDPs) between instructions.
- Architecture Items -- an overview of architecture building blocks, including:
  - Services, listing the name and description of associated services.
  - Data Objects (DOBs), listing the identifier (ID), name and description.
- Profiles -- All Profiles (PRFs) allocated to these Service Instructions, and for each starting with the description followed by a table with one or more group of one or more standards, indicating for each group the obligation (Mandatory, Conditional or Emerging) and the list of standards with their ID and title. This is followed by a table for usage, which lists all associated TINs with obligation and conditionality remarks.

5.1 Interactions

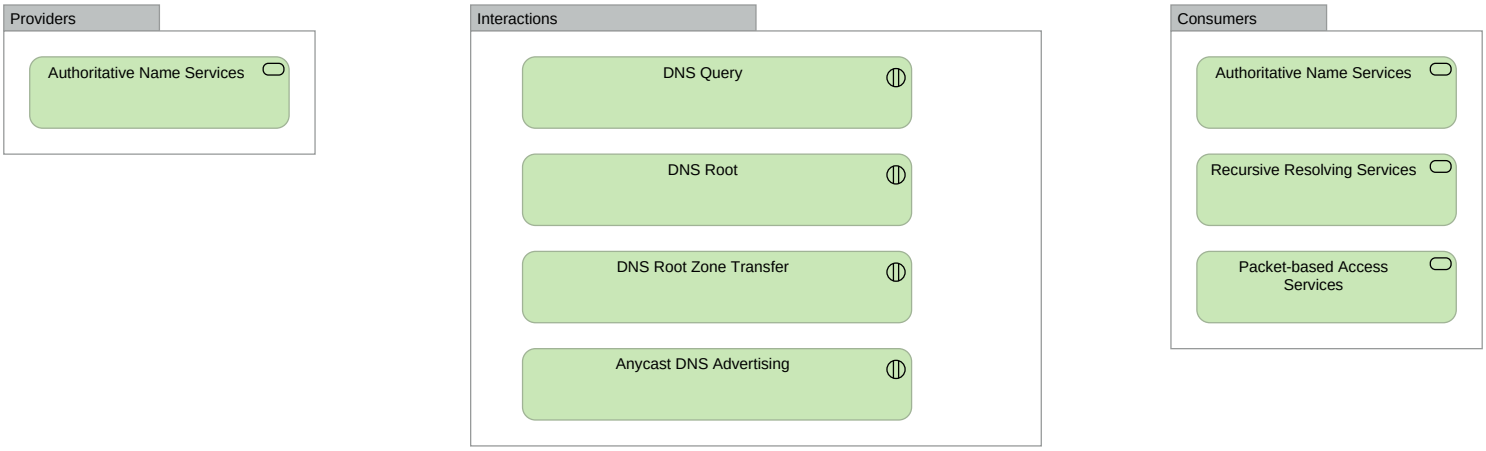

Figure 5. Interactions overview

5.1.1 DNS Query

(TIN-26) -- A DNS query is an exchange between a DNS resolver and an authoritative name server. The authoritative name server will answer the query according to the locally configured data, refer the resolver to a different authoritative DNS server of a delegated zone or else reject the query.

Overview

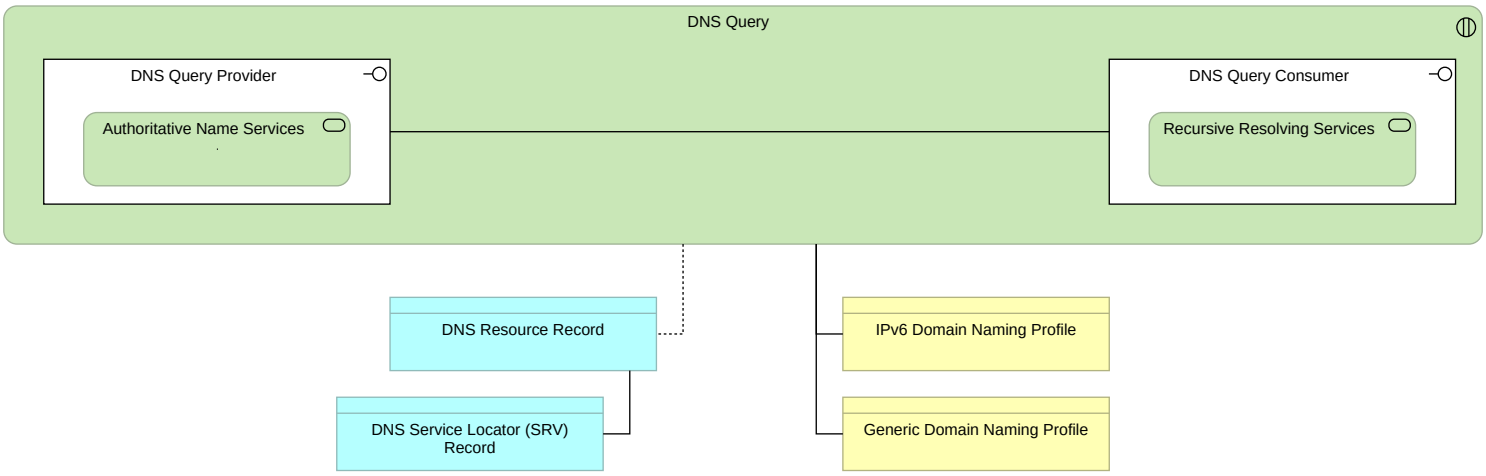

Figure 6. DNS Query Interaction

Secure Domain Naming

It is possible to inject invalid response for a DNS query. While many services use TLS security and refuse to communicate with a rogue host protecting confidentiality, forging responses allows a long-lasting denial-of-service attack, for example.

The Domain Name System Security Extensions (DNSSEC) provide a way to authenticate both existance and non-existance (optional) of DNS records.

DNSSEC is only used to validate records of DNS zone if and only if:

- Trust anchor exists;
- Each parent zone (up to the trust anchor) includes DS record for the child zone along to the NS child records;
- Resolving name server is configured to validate DNS Security Extensions and has the trust anchor.

If any of those conditions are not meet, then operation reverts to normal DNS operation. DNS Security Extensions (DNSSEC) are described in RFC 4033, RFC 4034, RFC 4035, RFC 4509, RFC 5155, and RFC 5702.

As FMN requires the root zone to be signed and delegation signers to be supported, each mission network participant can benefit from the additional DNSSEC secrity properties, if they they decide to provide signed zones and validating resolvers.

Keys

Each zone has two kind of keys: the Key Signing Key (KSK) that is used to sign Zone Signing Key (ZSK). The ZSK is used to sign the records in the zone.

The KSK can be stored off-line depending on required security level. The KSK has typical lifetime of 5 years in Internet use. KSK is used to sign ZSK that typically has lifetime of one year. When key roll-over takes place, both old and new key records are included zones over transition period, typically for one month.

The ZSK private key can also can be maintained off-line and only signed zone information will be stored on-line. This requires proper process to be set up sign zones and rotate keys when needed. Zones must be resign when they are modified, signatures expire or signing keys expire. Many DNS server software can take care of this automatically as an alternative. Automated zone signing is required if dynamic DNS updates are used.

Zones can be signed with different algorithms. Only the following security algorithms shall be used: RSASHA256, RSASHA512, ECDSAP256SHA256 and ECDSAP384SHA384.

Key Distribution

DNSSEC is required to be implemented from the root. The Mission Network PKI Manager will publish an trust anchor that is distributed. The mandatory way is to publish at least the DS records.

Each MNP will provide appopriate DS key records to the operator of the parent zone.

In case of non-root trust anchors, each sub-tree trust anchor must be distributed for all parties willing to verify DNS Security Extensions. Each party will need to configure their resolver to use those anchors.

Key and digest material have to be distributed securely. This includes the trust anchor (to all participating) and the DS records (to parent zones).

Partial Deployment

DNS Security Extensions allow partial deployment. Resolving name server may be configured not to verify zones even if a zone is signed. If the zone does not have DS records, the zone records won't be validated.

Example

Mission network has top level domain 'dnsmn'. The root and 'dnsmn' are signed and have associated key records. Two mission partners using mission TLD are 'ang' and 'hur'. In addition, one MNP wants to use their national ccTLD 'xd' under which it has mission-specific domain 'xmn'.

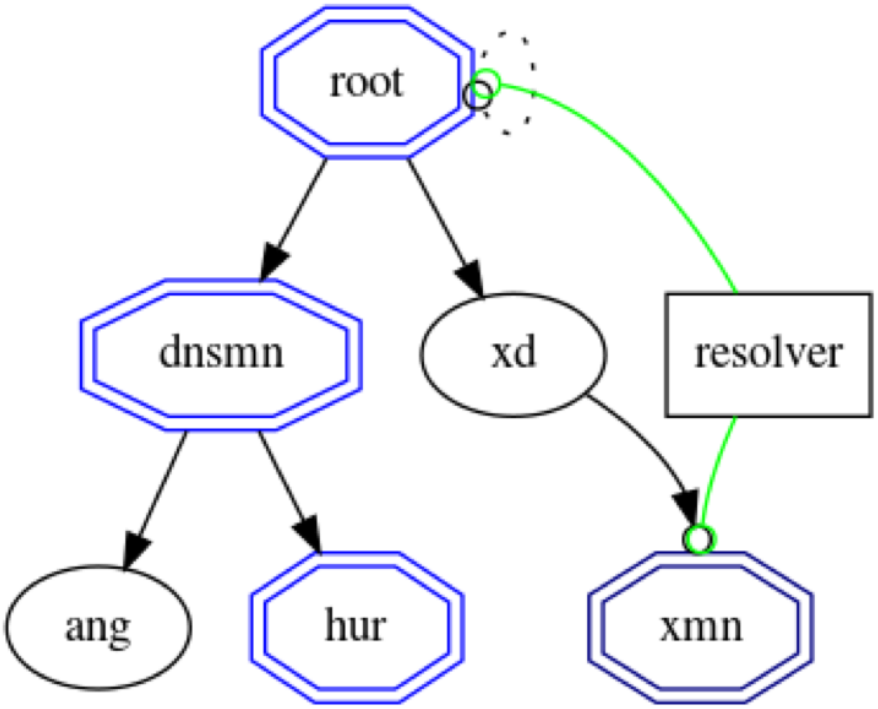

Figure 7. DNSSEC Example

The 'ang' does not want to deploy DNSSEC, so it does not add DNSSEC records to its zone and the DS records are not published in 'dnsmn' zone either. The 'hur' however, will hand out DS records for DNS administrator of 'dnsmn' and adds corresponding DS and DSKEY records to 'hur.dnsmn' zone.

When the trust anchor (the root key signing key public part) is configured into resolving server, that server is able to verify 'anything.hur.dnsmn' DNS records. For 'something.ang.dnsnm' verification is not done. Resource records are returned as without DNSSEC. However, as the parent zone is signed, also name server records of 'ang.dnsmn' are also signed and can be verified.

The 'xd' domain is not signed within MN as the real 'xd' will be in Internet. However, 'xmn.xd' provides a trust anchor. This can be configured to resolving name servers allowing validation of 'xmn.xd' domains.

Zone	Has DSKEY	Has TA	RR signed	NS records signed

root	Yes	Yes	Yes	Yes
dnsmn	Yes	No	Yes	Yes
hur.dnsmn	Yes	No	Yes	Yes
ang.dnsmn	No	No	No	Yes
xd	No	No	No	No
xmn.xd	Yes	Yes	Yes	No

Table 1. DNS Configuration

If the resoving server has not DNSSEC configured, the function is identical to the case where no DNSSEC records exists. No DNSSEC related records are transmitted in the network.

Implementation Considerations

While in many excercises second and third level domains (like mission.example) have been directly configured into root zone, this shall not be done in FMN missons and it may break DNSSEC. Accordingly, it is advisable to host intermediate zones that are also governed by the SMA (ip6.arpa, in-addr.arpa, arpa, mission TLD, etc.) on the same anycast services as the root zone itself.

The DNS Security Extensions protect authentity and integrity of DNS records betweeen authoritative name servers and resolving nameservers. It does not protect queries and responses between client and resolving nameserver. The client will know from a flag in the response if request was authenticated with DNSSEC.

Limitations

DNSSEC does not protect traffic confidentiality of queries.

Additional Resource Records

Additional resource records are needed.

- DS records defining digest of key signing keys for a zone
- DSKEY records defining zone keys, both key signing and zone signing keys
- NSEC and NSEC3 records allows authoritative name server to prove that a specific name does not exists.
- RRSIG cryptographically authenticates a resource record

Profiles

Profiles	Obligation	Conditionality
• Generic Domain Naming Profile (PRF-124)	Mandatory	
• IPv6 Domain Naming Profile (PRF-125)	Emerging	Networks with support of IPv6.

Requirements

Endpoint	Requirement	Dimension
Provider	The Domain Name Service shall support mission top level domains. (SREQ-234)	Technical
Provider	The Domain Name Service shall support country code top level domains. (SREQ-235)	Technical
Provider	The Domain Name Service must support, at a minimum, the SOA-record, NS-record and A-record matching to the NS-records. (SREQ-238)	Technical
Provider	The Domain Name Service must support the reverse zone containing the matching PTR-records. (SREQ-239)	Technical
Provider	The Authoritative Name Services shall allow delegation of reverse lookup zones of IPv4 prefixes that align on full 8 bits. (SREQ-614)	Technical
Provider	The Authoritative Name Services shall not support Recursive DNS Queries. (SREQ-616)	Technical
Provider	The Authoritative Name Services shall support the provision of two independent name servers for each delegated zone. (SREQ-613)	Technical
Provider	The Domain Name Service shall be able to delegate the signing of subdomains. (SREQ-380)	Technical
Provider	A distinct zone file shall be created for each signed zone. (SREQ-378)	Technical

Provider	The Authoritative Name Services shall be able to delegate unsigned zones. (SREQ-379)	Technical
Provider	The Domain Name Service shall be able to serve signed zones. (SREQ-382)	Technical
Provider	For name servers that have IP addresses listed as glue, the IP addresses must match the authoritative A records for that host. (SREQ-1311)	Technical
Provider	The set of NS records served by the authoritative name servers must match those proposed for the delegation in the parent zone. (SREQ-1312)	Technical
Provider	The data served by the authoritative name servers for the designated zone must be consistent. (SREQ-1313)	Technical
Provider	All authoritative name servers must serve the same NS record set for the designated domain. (SREQ-1314)	Technical
Provider	All authoritative name servers must serve the same SOA record for the designated domain. (SREQ-1315)	Technical
Provider	DNS response payload must not exceed 512 octets, where the delegation information in the referral is a complete set of NS records and one A record (SREQ-1316)	Technical
Provider	For any zones that are not anycasted, responses from the authoritative name servers must contain the same source IP address as the destination IP address of the initial query. (SREQ-1317)	Technical
Provider	Trust anchors must be provided as complete DS records, including the key tag, the key algorithm, the digest hash type, and the digest hash. (SREQ-1318)	Technical
Provider	At the time of the listing request there must be a DNSKEY present in the child zone that matches each DS record. (SREQ-1319)	Technical
Provider	Validation of the zone must be possible using the DS record set that has been provided for listing in the parent zone. (SREQ-1320)	Technical
Consumer	The Domain Name Service must support, at a minimum, the SOA-record, NS-record and A-record matching to the NS-records. (SREQ-238)	Technical
Consumer	The Domain Name Service shall operate without using forwarders for zones hosted by other MNPs. (SREQ-615)	Technical
Consumer	The Domain Name Service shall use a configured unicast IP-address when queries are made to other servers. (SREQ-611)	Technical

Configuration Parameters

ID	Name	Description
CFG-174	root.hints	Information on the root name servers needed to initialize resolvers. The field can hold the actual information or a reference to a location where the information may be found (i.e. URL)
CFG-179	Mission Network Domain Name Space	DNS name for mission use. Mission network participants may use delegation below the Mission Network Domain Name or mission-independent names.
CFG-227	**Trust Anchors**	List of zones and trusted keys for anchoring DNSSEC validation. While there may be multiple anchors, typically this list contains only the trust anchor for the root zone.
CFG-229	• Trusted Key	DNSSEC key that serves as trust anchor for the given zone in DNSKEY format.
CFG-228	• Zone Name	Name of the zone for which the trust anchor applies.
CFG-230	**List of Delegations**	The SMA has to keep a list of delegations, from which the root zone etc. are created.
CFG-172	• Delegated Zone	Zones to be delegated (forward and reverse).
CFG-231	• Authoritative Name Servers	List of Name Server that are authoritative for the given zone.
CFG-171	• Name Server	The fully qualified domain name (FQDN) of an authoritative name server for the name server (NS) entry.
CFG-170	• Name Server Addresses	The (IP) address(es) of the authoritative name server.
CFG-173	• Delegation Signers	Optional DS records to delegate DNSSEC signing keys to a zone. Signing keys shall be defined through both DS and DNSKEY record types
CFG-224	**Recursive Resolver Configuration**	Parameters and properties of the DNS Resolver(s)
CFG-175	• DNS Resolver Address	IP address of the recursive resolver, which clients can use to resolve domain names.
CFG-225	• DNSSEC Validation	Whether or not the Recursive Resolver performs DNSSEC validation.

Prerequisites

ID	Interaction	
	Obligation	Description / Conditionality
Service Instructions for Communications		
IDP-191	Communications Service Consumer Interface (TIN-112)	
	Mandatory	The Domain Name Services depend on the Communication services to query other federated Domain Name Servers over the network. Default QoS parameters for the DNS Query: • Application Type: Domain Name Server • Protocol: DNS • Transport Profile: TCP/UDP:53 • Service Class: SC3 • Precedence: Priority • DSCP Bits: 011110
Service Instructions for Distributed Time		
IDP-194	Federation Time Distribution (TIN-20)	
	Conditional	The Authoritative Name Services depend on time synchronization for timestamps in zones that are signed with DNSSEC. The Recursive Resolving Services depend on time synchronization to check timestamps included in signatures of resource records. **Conditionality** If zones shall be signed with DNSSEC or signed responses shall be validated
Service Instructions for Domain Naming		
IDP-222	Anycast DNS Advertising (TIN-114)	
	Conditional	The Authoritative Name Services may use Anycast DNS for a more robust service design. **Conditionality** Depending on mission service design
IDP-1118	DNS Root (TIN-370)	
	Mandatory	The Domain Name Services need a DNS Root to start recursive resolving.

5.1.2 DNS Root

(TIN-370) -- The DNS Root Technology Interaction is technically a DNS Query, but with some additional requirements on the authoritative name servers that host the root zone.

Conditionality

Authoritative name server serves the root zone.

Overview

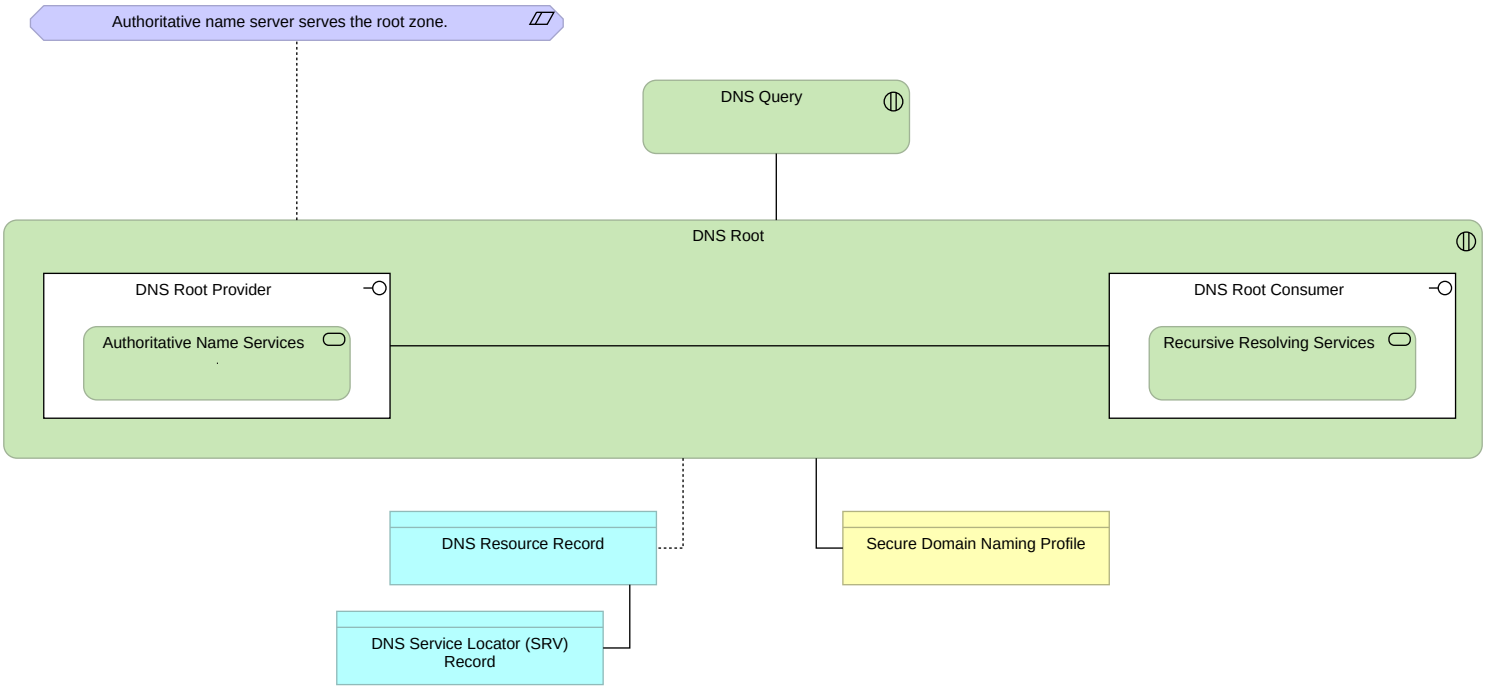

Figure 8. DNS Root Interaction

Profiles

Profiles	Obligation	Conditionality
• Secure Domain Naming Profile (PRF-80)	Mandatory	

Requirements

Endpoint	Requirement	Dimension
Provider	The Domain Name Service shall support mission top level domains. (SREQ-234)	Technical
Provider	The Domain Name Service shall support country code top level domains. (SREQ-235)	Technical
Provider	The Authoritative Name Services must support the ability to provide the root zone. (SREQ-242)	Technical
Provider	The Authoritative Name Services shall support the serving of the root zone by each MNP to its own resolvers (SREQ-617)	Technical
Provider	The root zone shall be signed. (SREQ-525)	Technical
Provider	The root zone shall be signed with one of the following algorithms: RSASHA256, RSASHA512, ECDSAP256SHA256, and ECDSAP384SHA384 (SREQ-1272)	Technical

Configuration Parameters

ID	Name	Description
CFG-226	**DNS Root**	Information about the DNS root, needed to bootstrap DNS resolving

CFG-174	• root.hints	Information on the root name servers needed to initialize resolvers. The field can hold the actual information or a reference to a location where the information may be found (i.g. URL)
CFG-227	• Trust Anchors	List of zones and trusted keys for anchoring DNSSEC validation. While there may be multiple anchors, typically this list contains only the trust anchor for the root zone.
CFG-229	• Trusted Key	DNSSEC key that serves as trust anchor for the given zone in DNSKEY format.
CFG-228	• Zone Name	Name of the zone for which the trust anchor applies.
CFG-178	DNS Root Zone	Root zone for the Domain Name Services, including refresh and retry intervals, expiry and minimum TTL, top-level domains and delegations.

5.1.3 DNS Root Zone Transfer

(TIN-43) -- A DNS Zone Transfer is a synchronization mechanism for authoritative DNS servers in which DNS zones are being transferred from a primary name server that hosts the original zone file to a secondary server that will host a copy.

In a federation context this may be the case, if a federation partner provides a secondary authoritative name server for a root or national domain. DNS Zone Transfer will also be used, if a mission partner manages it's own zone on a hidden primary server from where it is transferred to an (exposed) authoritative name server of a mission partner.

Conditionality

Servers hosting the root zone MUST support DNS Zone Transfer

Overview

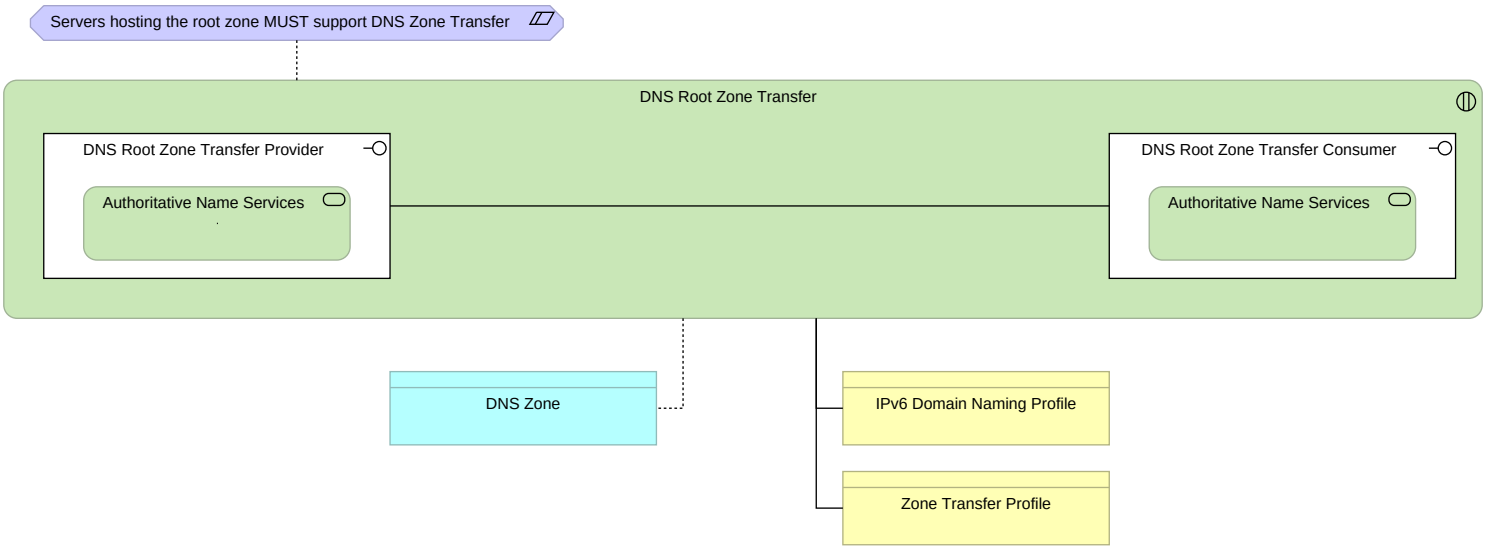

Figure 9. DNS Root Zone Transfer Interaction

Distribution Pattern

There are two modes of operation for the population of the zone file:

- A statically configured zone; and
- Zone updates via zone transfers.

Typically, an implementation will use a statically configured zone for the autonomous and ad-hoc capability.

While a mission deployment may become more permanent, zone transfers allow possibility to automatically distribute changes in the zones: Primary zone files will be created and configured on primary servers. The primay servers are then being used as the sources for zone transfers to the other authoritative servers that serve the same zone within the federation (secondary servers).

Zone Updates

Secondary authoritative servers will periodically, given by the root zone SOA retry timer, send requests for zone transfer (AXFR query) of the zone to the configured address(es) of the primary server(s). If a primary server is present in the federated network and responds to zone transfer requests, its zone data will be downloaded, keeping the distributed zone up to date in all servers.

Additional name servers in a federation are configured to use a Provider’s fellow name server as source for zone transfers. In order to update zone information (allow record updates) but to maintain standalone operation for extensive periods of time, zone transfers shall be adequately configured. This means that refresh timers should be relatively low (hours), while retention of data should be long (days to the months).

To further speed up convergence of a modified zone in the federation, the primary name server may be configured to send zone change notifications (NOTIFY) to the secondary servers which may trigger instant re-transfers after a change of the zone.

Hidden primary servers

Hidden primary servers are primary servers, which are the sources of zone transfers to other authoritve name servers, but which do not take part in normal DNS exchanges - i.e. they are not configured as a nameserver for a zone, nor do they answer to "normal" DNS queries.

As a hidden primary server needs only to be reachable by a well-defined set of secondary servers with a well-defined set of exchanges (AXFR/IXFR/NOTIFY) strict access control can be applied which enhances its availability, integrity and general security posture.

Profiles

Profiles	Obligation	Conditionality
• Zone Transfer Profile (PRF-122)	Mandatory	
• IPv6 Domain Naming Profile (PRF-125)	Emerging	Networks with support of IPv6.

Requirements

Endpoint	Requirement	Dimension
Provider	The Domain Name Service must support zone updates via zone transfers. (SREQ-241)	Technical
Provider	The Authoritative Name Services shall support being hidden (not accessible for normal DNS queries) when operating as a root server. (SREQ-612)	Technical
Provider	The Domain Name Service shall use a configured unicast IP-address when queries are made to other servers. (SREQ-611)	Technical
Provider	Authoritative Name Services shall secure DNS Zone Transfers with Secret Key Transaction Authentication for DNS (TSIG). (SREQ-529)	Technical
Consumer	The Domain Name Service must support zone updates via zone transfers. (SREQ-241)	Technical
Consumer	The Domain Name Service shall use a configured unicast IP-address when queries are made to other servers. (SREQ-611)	Technical
Consumer	Authoritative Name Services shall secure DNS Zone Transfers with Secret Key Transaction Authentication for DNS (TSIG). (SREQ-529)	Technical

Configuration Parameters

ID	Name	Description
CFG-220	**Zone Transfers**	Configuration of a DNS Zone Transfer
CFG-221	• Zone Name	Full name of the zone to be transfered
CFG-176	• Transfer Source	Address of the authoritative name server hosting the original zone file.
CFG-177	• Transfer Destination	Address of the authoritative name server that receives the zone file.
CFG-222	• Key name	TSIG NAME record - i.e. a mnemonic key name in domain name format
CFG-223	• Shared Secret	The key itself, which is used to authenticate the zone transfer

Prerequisites

ID	Interaction	
	Obligation	Description / Conditionality
Service Instructions for Communications		

IDP-220	Communications Service Consumer Interface (TIN-112)	
	Mandatory	The Domain Name Services depend on the Communication services to transfer DNS zones between federated Domain Name Servers over the network. Default QoS parameters for Zone Transfer: • Application Type: Domain Name Server • Protocol: DNS • Transport Profile: TCP:53 • Service Class: SC1 • Precedence: Priority • DSCP Bits: 001110

5.1.4 Anycast DNS Advertising

(TIN-114) -- Anycast is a process where same IP address is used in several entities providing the same service within network. In addition to have servers with same IP address, reachability to this address at particular server needs to be disseminated to the network. This is done with normal IP routing process. Shortest path calculation of forwarding tables for each destination is used to select the server in each point of network. This way traffic load to a service is distributed across the network, and also resiliency of service is increased when more than one server is responsible of that service. In context of DNS anycasting means that there can be several name servers having same IP address and providing authoritative response to the same zone. Use of the anycast for the DNS resolving process assumes that zone data between anycast servers is synchronized, and that IP network routing is used to direct queries to a server that has best ability to serve query.

IP anycast addressing can be used for the authoritative name servers to decouple the logical availability of the service from the network topology and the physical topology.

Conditionality

Anycast DNS MUST be supported by DNS providers that serve the root zone (root servers).

Overview

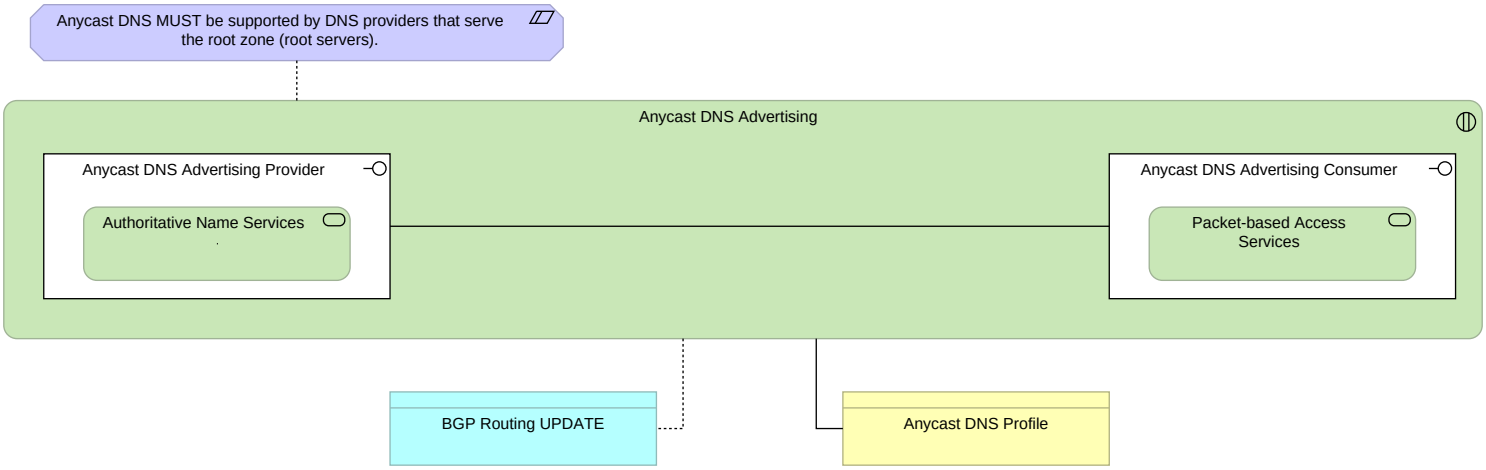

Figure 10. Anycast DNS Advertising Interaction

Implementation

Anycast service is implemented via dynamic routing updates of anycast service address based on name server ability to serve zone that anycast address is bound. This can be implemented via routing daemon that is embedded into the name server. A routing daemon announces host route to the name server when Authoritative Name Services for particular zone are activated. This way only zones that are served are announced to the mission network. Once there is a failure on the Authoritative Name Service or the name server no longer serves particular zone, anycast address associated to that zone is withdrawn by the routing daemon. This requires a control process within routing daemon that monitors name server ability to serve the zone-specific data. Implementation of this control is a Provider internal concern.

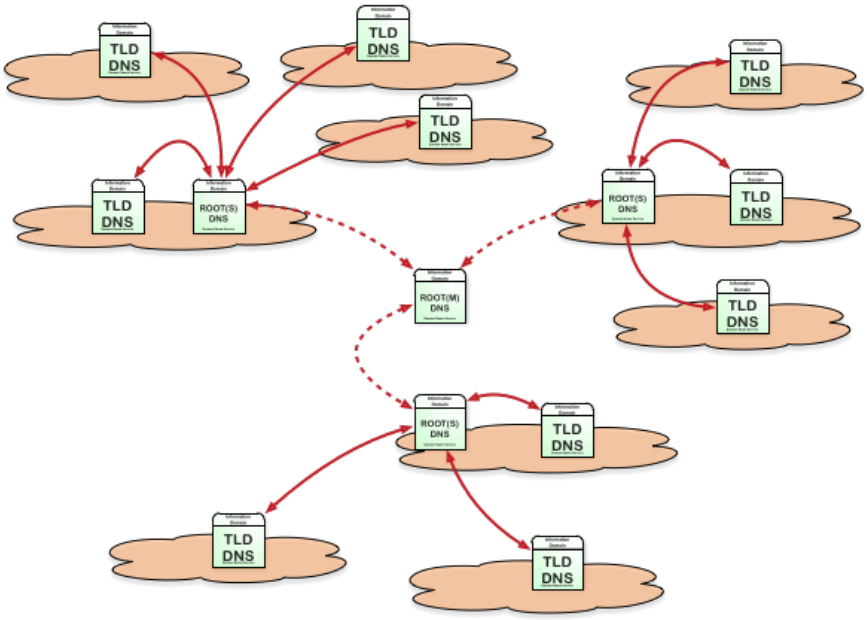

Figure 11. Anycast DNS Implementation

Current best practice to implement anycasting is to use node specific Autonomous System Numbers (ASN). This way all anycasted services will have not just specific addresses, but also specific AS number per node that anycasts the information. Allocation of these node specific anycast ASNs and rules that apply for the routing control are explained in the Service Instruction for Communications.

Anycasting Zones

Each DNS server that acts as anycast server for particular zone must have anycast address associated to it. Server uses this anycast address to listen zone specific queries, and responds to these queries via same anycast address. Server must also have unique unicast IP-address that is used when server does zone transfers and can also be used to monitor freshness of zone data across anycast servers.

An anycasted zone has name server (NS) delegations (and possibly glue records) that point to the anycast address in the delegating parent. This anycast address may then be served by any number of actual name servers as long as they have proper zone data. Root is a special case of anycast name server. It needs to have direct pointers in local resolvers "root.hints file". This will effectively direct queries to a nearest name server serving the root zone rather than a remote root. For autonomous operation of DNS structure it is recommended that each MN Provider is serving the root zone in addition to the local zones. This way potential connection issues within the mission network do not influence to the Domain Name Service. It is crucial for the availability of the zone that providers do not spread false information regarding the availability of zone data in its routing updates to neighboring Providers. When a Provider’s authoritative name server is unable to respond to DNS queries, its routers must no longer originate any anycast addresses corresponding to inaccessible zones. Such self-originated anycast routes must be withdrawn from all outbound updates to adjacent Providers.

This specification does not regulate the DNS client (stub resolver) interaction with the resolver because this is a Provider internal concern. Generally, clients find out which resolvers to use through DHCP or static configuration. It is however worth noticing that using anycast addresses for the resolver may also offer some advantages.

Security Implications

The fact that the root zone is replicated between participants and that the nearest server will be chosen based on anycast address, provides resilience against Distributed Denial-of-Service (DDoS) attacks. Attempted DDoS attacks will only affect the nearest authoritive servers. If the attacking systems are spread across the network, the induced load is balanced between several authoritive servers.

Profiles

Profiles	Obligation	Conditionality
• Anycast DNS Profile (PRF-123)	Mandatory	

Requirements

Endpoint	Requirement	Dimension
Provider	The Domain Name Service shall support anycast with a configurable address (SREQ-609)	Technical
Provider	The Domain Name Service shall respond the anycast queries using the same anycast address that it is listening on. (SREQ-610)	Technical
Provider	The Domain Name Service shall use a configured unicast IP-address when queries are made to other servers. (SREQ-611)	Technical

Configuration Parameters

ID	Name	Description
CFG-633	**Anycast DNS Service**	The anycast service configuration holds all all required parameters to configure Anycast DNS.
CFG-635	• Anycast ASN	The Anycast Autonomous System Number is the agreed ASN to be used for the Anycast DNS service.
CFG-634	• IP Address	The IP address(es) to be used for the Anycast DNS Service

Prerequisites

ID	Interaction	
	Obligation	Description / Conditionality
Service Instructions for Communications		
IDP-151	Communications Service Consumer Interface (TIN-112)	
	Mandatory	If the Authoritative Name Service is operating on a Shared Unicast Address, Anycast routes have to be announced to the Packet Routing Services and re-tracted automatically in case of local service degradation.
Service Instructions for Domain Naming		
IDP-223	DNS Root Zone Transfer (TIN-43)	
	Mandatory	Anycast DNS uses the DNS Zone Transfer to synchronize the served zone between anycast servers.

5.2 Architecture Items

5.2.1 Services

Name	Description
Authoritative Name Services	The Authoritative Name Services provide Domain Name Services data that is held in locally stored zones. A zone is the complete set of data available for a given node of the global DNS tree. Zone data can either be locally configured (manually or using some kind of automated process) or it can be synchronzized from a primary server in a process called "zone transfer".
Packet-based Access Services	The Packet-based Access Services provide the delivery or exchange of data (or digitized voice, video) encapsulated in IP packets.
Recursive Resolving Services	The Resolving Services perform the lookup part of the Domain Name Services in a process that interatively queries the Authoritative Name Services, beginning with the root servers, which serve the DNS root zone. Additionally to the plain resolving process, Resolving Services may offer additional functionality like caches that speed up the resolving process after the first query or validation the authenticity of the returned results with DNSSEC.

5.2.2 Data Objects

ID	Name	Description
DOB-14	DNS Resource Record	A resource record contains data conveyed by the Domain Name System. Typically this is an IP adress assigned to a domain name, but other record types are available too. *(RFC 1035)*
DOB-70	DNS Service Locator (SRV) Record	Generalized service location record, used for newer protocols instead of creating protocol-specific records such as MX. *(RFC 2782)*
DOB-16	DNS Zone	The DNS Zone encompasses all resource records assigned to a specific node in the DNS hierarchy. A ususal representation is the "zone file". *(RFC 1035)*
DOB-34	BGP Routing UPDATE	BGP routes are advertised between BGP speakers in UPDATE messages. Multiple routes that have the same path attributes can be advertised in a single UPDATE message by including multiple prefixes in the NLRI field of the UPDATE message. *(RFC 4271)*

5.3 Profiles

5.3.1 Generic Domain Naming Profile

(PRF-124) -- The Generic Domain Naming Profile provides base standards and guidance to support the hierarchical distributed name system for computers, services, or any resource connected to a federated mission network.

Obligation	Standards
Mandatory	Additional types and bigger payloads • RFC 2782 - "A DNS RR for specifying the location of services (DNS SRV)" • RFC 6891 - "Extension Mechanisms for DNS (EDNS(0))" • RFC 7766 - "DNS Transport over TCP - Implementation Requirements"
Mandatory	Base standards • RFC 1034 - "Domain names - concepts and facilities" • RFC 1035 - "Domain names - implementation and specification" • RFC 2181 - "Clarifications to the DNS Specification" • RFC 1123 - "Requirements for Internet Hosts - Application and Support"

Usage

Technology Interaction	Obligation	Conditionality
DNS Query (TIN-26)	Mandatory	

5.3.2 IPv6 Domain Naming Profile

(PRF-125) -- The IPv6 Domain Naming Profile contains additions to the base Domain Name System standards, which enable the usage of the Domain Name System in the context of the Internet Protocol, version 6.

Obligation	Standards
Emerging	DNS Extensions for IP version 6 *Conditionality* *IPv6 supported in network.* • RFC 3596 - "DNS Extensions to Support IP Version 6"
Emerging	Address Selection *Conditionality* *Resolvers and authoritative name servers that participate in zone transfers in networks that support IPv6.* • RFC 6724 - "Default Address Selection for Internet Protocol Version 6 (IPv6)"

Usage

Technology Interaction	Obligation	Conditionality
DNS Query (TIN-26)	Emerging	Networks with support of IPv6.
DNS Root Zone Transfer (TIN-43)	Emerging	Networks with support of IPv6.

5.3.3 Anycast DNS Profile

(PRF-123) -- The Anycast DNS Profile provides standards and guidance for operating an Authoritative Name Service on an anycast address.

Obligation	Standards
Mandatory	DNS operation on shared unicast address • RFC 3258 - "Distributing Authoritative Name Servers via Shared Unicast Addresses"

Mandatory	Operation of anycast services • RFC 4786 - "Operation of Anycast Services" • RFC 6382 - "Unique Origin Autonomous System Numbers (ASNs) per Node for Globally Anycasted Services"

Non-normative Standards

- RFC 7094 - Architectural Considerations of IP Anycast

Usage

Technology Interaction	Obligation	Conditionality
Anycast DNS Advertising (TIN-114)	Mandatory	

5.3.4 Zone Transfer Profile

(PRF-122) -- The Zone Transfer Profile provides standards and guidance to support zone synchronization in the hierarchical distributed name system for authoritative name servers of federated mission network ing.

Obligation	Standards
Mandatory	• RFC 1034 - "Domain names - concepts and facilities" • RFC 1035 - "Domain names - implementation and specification" • RFC 5936 - "DNS Zone Transfer Protocol (AXFR)"
Mandatory	Mandatory message digest algorithm is hmac-sha384. • RFC 8945 - "Secret Key Transaction Authentication for DNS (TSIG)"

Usage

Technology Interaction	Obligation	Conditionality
DNS Root Zone Transfer (TIN-43)	Mandatory	

5.3.5 Secure Domain Naming Profile

(PRF-80) -- The Secure Domain Naming Profile provides standards and guidance to support the hierarchical distributed name system with a set of extensions to DNS which provide to DNS clients (resolvers) cryptographic authentication of DNS data, authenticated denial of existence, and data integrity, but not availability or confidentiality. These extensions are combined in the Domain Name System Security Extensions (DNSSEC), a suite of Internet Engineering Task Force (IETF) specifications for securing certain kinds of information provided by the Domain Name System (DNS) as used on Internet Protocol (IP) networks.

Obligation	Standards
Mandatory	• RFC 4033 - "DNS Security Introduction and Requirements" • RFC 4034 - "Resource Records for the DNS Security Extensions" • RFC 4035 - "Protocol Modifications for the DNS Security Extensions" • RFC 4509 - "Use of SHA-256 in DNSSEC Delegation Signer (DS) Resource Records (RRs)" • RFC 5155 - "DNS Security (DNSSEC) Hashed Authenticated Denial of Existence" • RFC 5702 - "Use of SHA-2 Algorithms with RSA in DNSKEY and RRSIG Resource Records for DNSSEC"

Implementation Scope

Only the following security algorithms shall be used:

- RSASHA256,
- RSASHA512,
- ECDSAP256SHA256,
- ECDSAP384SHA384.

Usage

Technology Interaction	Obligation	Conditionality
DNS Root (TIN-370)	Mandatory	

6 Procedures

The execution of activities in support of these service instructions is captured in business process descriptions. Each process description includes the performance of activities in pre-defined roles.

This chapter shows all business processes (BPs) associated with the service instructions with their respective description and a table that lists the subordinate Business Activities (BAs), including their identifier (ID), name and definition of that activity, and the Business Role (ROLE) that performs the activity for that process in the context of these instructions.

6.1 Planning Procedures

6.1.1 Federated Domain Name Services Framework Process

(BP-453) -- The Federated Domain Name Services Framework Process encompasses activities that are performed by Affiliates outside of a concrete mission planning process, but which set foundations for future mission instantiations

Activities

Process Steps		
ID	Activity	Role
BA-8	*Acquire national domain name space* Request national domain space from an Internet Registry, if required.	Authoritative Name Service Provider

6.1.2 Federated Domain Name Services Design Process

(BP-454) -- The Federated Domain Name Service Design Process encompasses planning and service design activities for a concrete mission instatiation.

Activities

Process Steps		
ID	Activity	Role
BA-1502	*Define mission name spaces* Define name space(s) for mission usage.	Mission Network Service Management Authority
BA-14	*Request domain name spaces for mission usage* Request mission domain name space from Mission Network Service Management Authority, or register national domain name space.	Authoritative Name Service Provider
BA-7	*Service Design for Domain Name Service* Define root servers; Decide on the usage of anycast and direct forward or stub zone resolving (Protected Core); Prepare root.hints file.	Mission Network Service Management Authority

6.1.3 Federated Domain Name Services Setup Process

(BP-300) -- The Federated Domain Name Service Setup Process includes preparations for a mission deployment that take place after the service has been designed, but before the actual Joining phase begins.

Activities

Process Steps		
ID	Activity	Role
BA-1503	*Generate Root Key* Generate Root Key Signing Keys and Zone Signing Keys; Prepare trusted keys file.	Mission Network Service Management Authority
BA-1504	*Prepare Root Zone* Generate and sign the root zone and other centrally managed zones (ip6.arpa etc.).	Mission Network Service Management Authority

Requirements

ID	Activity	Requirements
BA-1504	Prepare Root Zone	• The root zone shall only delegate top level domains (TLD). (SREQ-1159)

6.2 Joining Procedures

6.2.1 Federated Domain Name Services Joining Process

(BP-299) -- The Federated Domain Name Services Joining Process establishes the Domain Name Services in a Federated Mission Network.

Activities

Process Steps		
ID	Activity	Role
BA-1505	*Install root.hints and trusted keys* Install root.hints and trusted keys (if validation is desired) into recursive resolvers.	Resolving Service Provider
BA-1506	*Request delegations* Provide delegated zones, domain name server addresses, names and delegation signers to Mission Network Service Management Authority.	Authoritative Name Service Provider
BA-4	*Delegate Zones* Configure requested delegations into centrally managed zones.	Mission Network Service Management Authority

6.3 Membership Procedures

6.3.1 Federated Domain Name Services Membership Process

(BP-301) -- The Federated Domain Name Services Membership Process maintains the Domain Name Services in a Federated Mission Network.

Activities

Process Steps		
ID	Activity	Role
BA-9	*Provide Authoritative Name Service* Operate authoritative name server and manage and maintain delegated zones.	Authoritative Name Service Provider
BA-11	*Provide Resolving Service* Operate a Recursive Resolver and provide Resolving Services to Service Consumers.	Resolving Service Provider

6.3.2 Federated Domain Name Services Change Management Process

(BP-457) -- The Federated Domain Name Services Change Management Process describes activities required to implement changes in the Domain Name Service.

Activities

Process Steps		
ID	Activity	Role
BA-1510	*Request changes to delegated zones* If required, request changes to delegated zones at the Authoritative Name Service provider.	Service Consumer
BA-1511	*Update delegated zones* Configure necessary changes in delegated zones.	Authoritative Name Service Provider

BA-1509	*Request changes to centrally managed zones* If required, request changes to centrally managed zones at Mission Network Service Management Authority.	Authoritative Name Service Provider
BA-1	*Update centrally managed zones* Configure necessary changes in centrally managed zones.	Mission Network Service Management Authority

6.4 Transfer of SMA Procedures

6.4.1 Federated Domain Name Services SMA Transfer Process

(BP-455) -- The Federated Domain Name Services SMA Transfer Process describes the activities required, if a central role of the Mission Domain Name Services changes from one provider to another.

Activities

Process Steps		
ID	Activity	Role
BA-17	*Generate and publish new Root Key* Generate new Root Keys, publish them in root zone; Distribute new trusted keys (and root.hints) files.	Mission Network Service Management Authority
BA-1507	*Update root.hints and trusted keys* Install new root.hints and trusted keys into recursive resolvers	Resolving Service Provider

6.5 Exit Procedures

6.5.1 Federated Domain Name Services Exiting Process

(BP-456) -- The Federated Domain Name Services Exiting Process describes the activities that have to be performed, if participant leaves a mission network, or the mission network is decommissioned.

Activities

Process Steps		
ID	Activity	Role
BA-1508	*Release delegations* Request reseasing of delegated zones at Mission Network Service Management Authority.	Authoritative Name Service Provider

Annex A. Definitions

Term	Definition
Authoritative Name Server	A name server that is authoritative for a specific zone.
Coloured Cloud	A Coloured Cloud (CC) is a cloud of a specific information (security) domain. If information confidentiality for the traffic transported between CCs is needed, crypto devices (Z) shall be used to tunnel traffic via the Protected Core (PCore). In such a case, these crypto devices (Z) are part of the corresponding CCs. (AComP-5637 Edition A Version 1)
Domain	A subtree of the domain name space under a Domain Name System (DNS) node.
Domain Name	The name of a node in the Domain Name System (DNS) hierarchical namespace expressed in a dotted notation. It consists of one or more parts, technically called labels that are conventionally concatenated, and delimited by dots, such as ‘nato.int’. The right-most label conveys the top-level domain; for example, the domain name ‘www.nato.int’ belongs to the top-level domain ‘int’, with ‘nato’ being a sub domain of the ‘int’ domain. (RFC 1035, RFC 1123 and RFC 2181)
Domain Name Resolution	Operation that takes a domain name and returns the corresponding IP-address(es) of a networked entity/device.
Domain Name Space	Tree structured name space and data associated with the names. Conceptually, each node and leaf of the domain name space tree names a set of information, and query operations are attempts to extract specific types of information from a particular set. (RFC 1034)
Domain Name System	Hierarchical distributed naming system mainly used for the translation of domain names of networked devices, services and resources in the corresponding IP address(es).
Mission Network	Single governed capability, including the communication and information systems, management, processes and procedures created for the purposes of an operation, exercise, training event, and/or interoperability verification activity, using a flexible and tailored set of non-materiel (policy, processes, procedures and standards) and materiel (static and deployed networks, services, supporting infrastructures) contributions provided by NATO, NATO Partners and non-NATO nations and entities participating in operations. (NATO FMN Implementation Plan, Volume I, Annex B, 23 September 2014)
Name Server	A machine that runs the Domain Name System (DNS) services and maintains the DNS database information about the DNS domain. When queried it provides either the requested information, a pointer to another DNS server that can further resolve the query, or a response message indicating that it does not have the information or that the information does not exist. A name server can be authoritative for one or more zones.
Node	Node in the domain name space tree.
PCore	The Protected Core (PCore) is a set of Protected Core Segments (PCSs) working together in a federation-of-systems approach to collectively achieve the characteristics of Protected Core Networking (PCN). That is, PCore is the overall core federated network offering protected infrastructure sharing, the elementary network elements of which are PCSs. (AComP-5637 Edition A Version 1)
Protected Core Networking	Protected Core Networking (PCN) is a concept used to establish a flexible but secure military transport infrastructure to support military operations offering high IP transport availability, efficient resource sharing, resilience and defence against cyber attacks. (AComP-5637 Edition A Version 1)
Protected Core Segment	A Protected Core Segment (PCS) is a network built on the principles of Protected Core Networking (PCN) and designed to work seamlessly with other PCSs in a federation of systems. Each PCS shall satisfy a set of functional requirements, called PCS characteristics, to support the PCN concept. (AComP-5637 Edition A Version 1)
Resolver	Generic code, program, or service that extracts information from name servers in response to client requests. Resolvers must be able to access at least one name server and use that name server's information to answer a query directly, or pursue the using referrals to other name servers.
Reverse Domain Name Lookup	See Reverse Domain Name Resolution. (Redirect)
Reverse Domain Name Resolution	Operation that converts IP address in the corresponding domain names.
Root Domain Name	The root domain name of a Domain Name System (DNS).
Root Name Server	The authoritative server for the root domain name.
Top Level Domain	One of the domains at the highest level in the hierarchical Domain Name System (DNS) of the Internet. The top-level domain names are installed in the root zone of the name space.
Zone	A subtree of the domain name space for which a name server is authoritative.

Annex B. Requirements

This annex provides an overview of requirements that are fulfilled with these service instructions, together with the specific aspects in which this has been achieved.

In the context of the service instructions there are two different levels identified: the requirements that are fulfilled at the level of the federation and those for the enterprise.

- For federation-level requirements, this annex shows the requirement statement per endpoint (Provider, Consumer or Mediator) and with the dimension (Process, People, Technical or Security), grouped per Technology Interaction (TIN).
- For enterprise-level requirements, the annex shows the requirement statement with ID and tile, together with the dimension (as above). In some cases, the requirements that are subordinate to a broader requirement are listed underneath in a bullet list.

Requirement statements will use the keywords "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL", which are to be interpreted as described in RFC 2119 "Key words for use in RFCs to Indicate Requirement Levels".

B.1 Federation-level Requirements

Interaction		
Endpoint	Requirement	Dimension
1. DNS Query (TIN-26)		
Provider	The Domain Name Service shall support mission top level domains. (SREQ-234)	Technical
Provider	The Domain Name Service shall support country code top level domains. (SREQ-235)	Technical
Provider	The Domain Name Service must support, at a minimum, the SOA-record, NS-record and A-record matching to the NS-records. (SREQ-238)	Technical
Provider	The Domain Name Service must support the reverse zone containing the matching PTR-records. (SREQ-239)	Technical
Provider	The Authoritative Name Services shall allow delegation of reverse lookup zones of IPv4 prefixes that align on full 8 bits. (SREQ-614)	Technical
Provider	The Authoritative Name Services shall not support Recursive DNS Queries. (SREQ-616)	Technical
Provider	The Authoritative Name Services shall support the provision of two independent name servers for each delegated zone. (SREQ-613)	Technical
Provider	The Domain Name Service shall be able to delegate the signing of subdomains. (SREQ-380)	Technical
Provider	A distinct zone file shall be created for each signed zone. (SREQ-378)	Technical
Provider	The Authoritative Name Services shall be able to delegate unsigned zones. (SREQ-379)	Technical
Provider	The Domain Name Service shall be able to serve signed zones. (SREQ-382)	Technical
Provider	For name servers that have IP addresses listed as glue, the IP addresses must match the authoritative A records for that host. (SREQ-1311)	Technical
Provider	The set of NS records served by the authoritative name servers must match those proposed for the delegation in the parent zone. (SREQ-1312)	Technical
Provider	The data served by the authoritative name servers for the designated zone must be consistent. (SREQ-1313)	Technical
Provider	All authoritative name servers must serve the same NS record set for the designated domain. (SREQ-1314)	Technical
Provider	All authoritative name servers must serve the same SOA record for the designated domain. (SREQ-1315)	Technical
Provider	DNS response payload must not exceed 512 octets, where the delegation information in the referral is a complete set of NS records and one A record (SREQ-1316)	Technical
Provider	For any zones that are not anycasted, responses from the authoritative name servers must contain the same source IP address as the destination IP address of the initial query. (SREQ-1317)	Technical
Provider	Trust anchors must be provided as complete DS records, including the key tag, the key algorithm, the digest hash type, and the digest hash. (SREQ-1318)	Technical

Provider	At the time of the listing request there must be a DNSKEY present in the child zone that matches each DS record. (SREQ-1319)	Technical
Provider	Validation of the zone must be possible using the DS record set that has been provided for listing in the parent zone. (SREQ-1320)	Technical
Consumer	The Domain Name Service must support, at a minimum, the SOA-record, NS-record and A-record matching to the NS-records. (SREQ-238)	Technical
Consumer	The Domain Name Service shall operate without using forwarders for zones hosted by other MNPs. (SREQ-615)	Technical
Consumer	The Domain Name Service shall use a configured unicast IP-address when queries are made to other servers. (SREQ-611)	Technical
3. DNS Root (TIN-370)		
Provider	The Domain Name Service shall support mission top level domains. (SREQ-234)	Technical
Provider	The Domain Name Service shall support country code top level domains. (SREQ-235)	Technical
Provider	The Authoritative Name Services must support the ability to provide the root zone. (SREQ-242)	Technical
Provider	The Authoritative Name Services shall support the serving of the root zone by each MNP to its own resolvers (SREQ-617)	Technical
Provider	The root zone shall be signed. (SREQ-525)	Technical
Provider	The root zone shall be signed with one of the following algorithms: RSASHA256, RSASHA512, ECDSAP256SHA256, and ECDSAP384SHA384 (SREQ-1272)	Technical
4. DNS Root Zone Transfer (TIN-43)		
Provider	The Domain Name Service must support zone updates via zone transfers. (SREQ-241)	Technical
Provider	The Authoritative Name Services shall support being hidden (not accessible for normal DNS queries) when operating as a root server. (SREQ-612)	Technical
Provider	The Domain Name Service shall use a configured unicast IP-address when queries are made to other servers. (SREQ-611)	Technical
Provider	Authoritative Name Services shall secure DNS Zone Transfers with Secret Key Transaction Authentication for DNS (TSIG). (SREQ-529)	Technical
Consumer	The Domain Name Service must support zone updates via zone transfers. (SREQ-241)	Technical
Consumer	The Domain Name Service shall use a configured unicast IP-address when queries are made to other servers. (SREQ-611)	Technical
Consumer	Authoritative Name Services shall secure DNS Zone Transfers with Secret Key Transaction Authentication for DNS (TSIG). (SREQ-529)	Technical
5. Anycast DNS Advertising (TIN-114)		
Provider	The Domain Name Service shall support anycast with a configurable address (SREQ-609)	Technical
Provider	The Domain Name Service shall respond the anycast queries using the same anycast address that it is listening on. (SREQ-610)	Technical
Provider	The Domain Name Service shall use a configured unicast IP-address when queries are made to other servers. (SREQ-611)	Technical

B.2 Enterprise-level Requirements

ID	Requirement	Dimension
SREQ-53	Events shall be logged by DNS servers and resolvers.	Security
SREQ-54	Event logs must register at least: zone transfers; failed requests; and zone modifications.	Security
SREQ-309	Event logs shall include details of failed requests.	Security
SREQ-310	Event logs shall include details of DNSSEC verification failures.	Security
SREQ-240	The Authoritative Name Services must support statically configured zones.	Technical

Annex C. Dependencies

This annex introduces the notion of dependency between the instructions in this spiral specification. Fact is that all instructions, both procedural and service, are somehow related to several other instructions in order to complete a process, deliver a product, or provide a service. These relations are identified as dependencies.

In the context of the service instructions there are two different categories identified: the relations from these service instructions depending on other ("prerequisites") and those from other depending on this service instructions ("usage").

- For prerequisites, this annex lists the dependencies in tables per applicable Technology Interactions (TINs) in the service instructions. The table shows the Technology Dependencies (IDPs), grouped per dependent instructions, each linked to the corresponding TINs of those other instructions, and stating the obligation (Mandatory or Conditional) with a description and a conditionality statement if available.
- For usage, the annex lists the dependencies also in tables, but separated into two distinct groups: those that cover supported Information Exchange Requirements (IERs) in depending procedural instructions, and those covering supported services in depending service instructions. The tables for the services are the same as above; those for the IERS link to IERS and express the obligation differently (Primary or Alternative).

C.1 Prerequisites

DNS Query (TIN-26)

ID	Interaction	
	Obligation	Description / Conditionality
Service Instructions for Communications		
IDP-191	Communications Service Consumer Interface (TIN-112)	
	Mandatory	The Domain Name Services depend on the Communication services to query other federated Domain Name Servers over the network. Default QoS parameters for the DNS Query: • Application Type: Domain Name Server • Protocol: DNS • Transport Profile: TCP/UDP:53 • Service Class: SC3 • Precedence: Priority • DSCP Bits: 011110
Service Instructions for Distributed Time		
IDP-194	Federation Time Distribution (TIN-20)	
	Conditional	The Authoritative Name Services depend on time synchronization for timestamps in zones that are signed with DNSSEC. The Recursive Resolving Services depend on time synchronization to check timestamps included in signatures of resource records. **Conditionality** If zones shall be signed with DNSSEC or signed responses shall be validated
Service Instructions for Domain Naming		
IDP-222	Anycast DNS Advertising (TIN-114)	
	Conditional	The Authoritative Name Services may use Anycast DNS for a more robust service design. **Conditionality** Depending on mission service design
IDP-1118	DNS Root (TIN-370)	
	Mandatory	The Domain Name Services need a DNS Root to start recursive resolving.

DNS Root Zone Transfer (TIN-43)

ID	Interaction	
	Obligation	Description / Conditionality
Service Instructions for Communications		
IDP-220	Communications Service Consumer Interface (TIN-112)	
	Mandatory	The Domain Name Services depend on the Communication services to transfer DNS zones between federated Domain Name Servers over the network. Default QoS parameters for Zone Transfer: • Application Type: Domain Name Server • Protocol: DNS • Transport Profile: TCP:53 • Service Class: SC1 • Precedence: Priority • DSCP Bits: 001110

Anycast DNS Advertising (TIN-114)

ID	Interaction	
	Obligation	Description / Conditionality
Service Instructions for Communications		
IDP-151	Communications Service Consumer Interface (TIN-112)	
	Mandatory	If the Authoritative Name Service is operating on a Shared Unicast Address, Anycast routes have to be announced to the Packet Routing Services and re-tracted automatically in case of local service degradation.
Service Instructions for Domain Naming		
IDP-223	DNS Root Zone Transfer (TIN-43)	
	Mandatory	Anycast DNS uses the DNS Zone Transfer to synchronize the served zone between anycast servers.

C.2 Usage

C.2.1 Supported Services

DNS Query (TIN-26)

ID	Interaction	
	Obligation	Description / Conditionality
Service Instructions for Data Links		
IDP-1101	JREAP Point to Point (TIN-108)	
	Mandatory	Data link services depend on the Domain Name System (DNS) services to locate services with which they federate. Therefore, the DNS services shall be operational and host names resolvable.
Service Instructions for Digital Certificates		
IDP-270	Digital Certificate Validation (TIN-165)	
	Mandatory	The Digital Certificates Services depend on the Domain Naming System (DNS) services to locate federated Digital Certificates Services on the network. Therefore the DNS services shall be operational and host names resolvable. The Digital Certificates Services servers DNS names shall be available to the contributing MN Participants.
Service Instructions for Friendly Force Tracking		
IDP-275	Consume FFT Tracks (TIN-76)	
	Mandatory	It is recommended to use DNS services to locate federated Friendly Force Tracking Services on the network. Hence the domain name services shall be operational and host names resolvable. The Friendly Force Tracking servers' DNS names shall be available to the contributing MN participants.
IDP-289	FFI MTF Exchange (TIN-169)	
	Mandatory	It is recommended to use DNS services to locate federated Friendly Force Tracking Services on the network. Hence the domain name services shall be operational and host names resolvable. The Friendly Force Tracking servers' DNS names shall be available to the contributing MN participants.
Service Instructions for Geospatial Information		
IDP-969	Geospatial Web Services (TIN-153)	
	Mandatory	The Geospatial Web Services depend on the Domain Name System (DNS) services to locate federated Geospatial Web Services on the network. Therefore the DNS services shall be operational and host names resolvable. The Geospatial Web Services servers DNS names shall be available to the contributing participants.
Service Instructions for Informal Messaging		
IDP-1276	Email routing (TIN-11)	
	Mandatory	Required for publication of MX records.
Service Instructions for Intelligence Battlespace Objects Exchange		
IDP-1124	Intelligence BsO Synchronization (TIN-371)	
	Mandatory	The Intelligence BsO Synchronization Service depend on the Domain Name System (DNS) services to locate services with which they federate. Therefore, the DNS services shall be operational and host names resolvable.
Service Instructions for Joint ISR Information Exchange		
IDP-558	Exchange ISR Library data and metadata (TIN-307)	
	Mandatory	The ISR Library services depend on the Domain Name System (DNS) services to locate services with which they federate. Therefore, the DNS services shall be operational and host names resolvable. To make CSD servers' DNS names available to MN participants, providers of Coalition Shared Data (CSD) servers SHALL register 'A' and "PTR" records for each CSD-server.
IDP-563	ISR Streaming Services Access (TIN-308)	
	Mandatory	The ISR Streaming services depend on the Domain Name System (DNS) services to locate services with which they federate. Therefore, the DNS services shall be operational and host names resolvable. To make CSD servers' DNS names available to MN participants, providers of Coalition Shared Data (CSD) servers SHALL register 'A' and "PTR" records for each CSD-server.
Service Instructions for Land C2 Information Exchange		

IDP-1070	Exchange BattleSpace Objects (TIN-362)	
	Conditional	The Situational Awareness Services and Battlespace Object Services implicitly depend on the Domain Name System (DNS) services when MIP4 EndPoints are expressed using domain names. **Conditionality** When MIP4 EndPoints are expressed using domain names.
Service Instructions for Protected Core Networking		
IDP-163	NMCD Service Configuration Information Exchange (TIN-115)	
	Mandatory	NMCD Information Exchange Service is depend on ability to resolve names
IDP-990	PCN NMCD Domain Name Service (TIN-352)	
	Mandatory	The federated NMCD management services may use domain name service to resolve service names to IP addresses and vice versa. Note that if root name server functionality is used within the Mission Network Protected Core NMCD federation, the use of DNSSEC on the root name servers is not mandated.
IDP-159	PCSOP Information Exchange (TIN-116)	
	Mandatory	NMCD PCSOP Information Exchange Service is depend on ability to resolve names
Service Instructions for Service Management and Control		
IDP-217	Problem Management API - Unregister Problem Listener (TIN-135)	
	Mandatory	XMPP SRV records iaw RFC 6120, RFC 6121
Service Instructions for Text-based Collaboration		
IDP-210	Instant Messaging (TIN-138)	
	Mandatory	The Text-based Collaboration Services depend on the Domain Name System (DNS) infrastructure for location of the correct servers. For this reason the DNS infrastructure configuration is preparatory to the Text-based Collaboration Services infrastructure configuration. All XMPP services must be advertised using DNS SRV records. As a minimum, each XMPP service is required to have: • A record (or CNAME) with the XMPP domain [XMPPDomain] • TCP SRV Record for _xmpp-server under [XMPPDomain] • TCP SRV Record for _xmpp-client under [XMPPDomain] • A record (or CNAME) with the conference service under XMPP domain [conference.XMPPDomain] Additional records may be needed for specific products and/or extensions. See the FMN Best Practices (Instantiation section) for further details. Connection endpoints must be reachable from remote locations to ensure correct federation of the service with other affiliates. As minimum, Affiliates should be able to reach TCP ports declared in the DNS SRV records for _xmpp-server and MN participants' users should be able to reach TCP ports declared in the DNS SRV records for _xmpp-client.
IDP-1277	Instant Messaging and Presence (TIN-369)	
	Mandatory	Required for publication of MX records.
Service Instructions for Web Authentication		
IDP-768	Web Authentication (TIN-64)	
	Mandatory	Web Authentication depends on the Domain Name System for locating Identity Providers and Relying Parties in the Mission Network. Therefore, the DNS services shall be operational and host names resolvable.
Service Instructions for Web Hosting		
IDP-641	Web-based Content Hosting (TIN-7)	
	Mandatory	The Web Hosting Services depend on the Domain Name Services to locate Web Hosting Servers. Therefore Domain Name Services shall be operational and host names resolvable. The Web Hosting Service server domain names shall be registered and available to the contributing participants.

DNS Root (TIN-370)

ID	Interaction	
	Obligation	Description / Conditionality
Service Instructions for Domain Naming		

IDP-1118	DNS Query (TIN-26)	
	Mandatory	The Domain Name Services need a DNS Root to start recursive resolving.

DNS Root Zone Transfer (TIN-43)

ID	Interaction	
	Obligation	Description / Conditionality
Service Instructions for Domain Naming		
IDP-223	Anycast DNS Advertising (TIN-114)	
	Mandatory	Anycast DNS uses the DNS Zone Transfer to synchronize the served zone between anycast servers.

Anycast DNS Advertising (TIN-114)

ID	Interaction	
	Obligation	Description / Conditionality
Service Instructions for Domain Naming		
IDP-222	DNS Query (TIN-26)	
	Conditional	The Authoritative Name Services may use Anycast DNS for a more robust service design. **Conditionality** Depending on mission service design
Service Instructions for Protected Core Networking		
IDP-988	PCN NMCD Domain Name Service (TIN-352)	
	Conditional	To improve the resiliency of name resolution in the NMCD federation, the domain name service may use the anycast DNS functionality to replicate root-level information to several PCSes in the federation. Additionally, mission network participant PCSes may anycast additional zones to improve their resiliency. **Conditionality** Anycast DNS functionality is used within the Mission Network Protected Core NMCD federation.

Annex D. Roles and Responsibilities

This annex provides information about the Business Roles (ROLE) that are performed to execute the various Business Activities (BAs) related to these service instructions. The organization of these activities as part of the Business Processes (BPs) is explained in the Context chapter.

The annex provides a table that provides all activities associated with a certain role. The roles are identified with their title and ID, sorted alphabetically by title. The second column provides the definition of the role and below that, the respective Business Processes in which the role is performed, with the specific Business Activities in a bullet list below.

Role	Definition and Activities
Authoritative Name Service Provider (ROLE-386)	The Authoritative Name Service Provider is the operator of an authoritative DNS server, which hosts one or multiple DNS zones.
	Activities *Federated Domain Name Services Membership Process (BP-301)* • Provide Authoritative Name Service *Federated Domain Name Services Design Process (BP-454)* • Request domain name spaces for mission usage *Federated Domain Name Services Joining Process (BP-299)* • Request delegations *Federated Domain Name Services Change Management Process (BP-457)* • Update delegated zones • Request changes to centrally managed zones *Federated Domain Name Services Framework Process (BP-453)* • Acquire national domain name space *Federated Domain Name Services Exiting Process (BP-456)* • Release delegations
Mission Network Service Management Authority (ROLE-22)	A Mission Network Service Management Authority (MNSMA) is a central role in a Mission Network assigned by the Lead Commander based on delegated authority from all mission network participants, to define and govern the ICT services within a mission. This includes the accountability for the design, provision, management, security, and provision of oversight and control of Information and Communications Services in a coherent, effective and coordinated manner within its designated Area Of Responsibility (AOR), and the assurance of the availability of ADS and TDS. The MNSMA establishes a standardized Service Management Framework (SMF) for the MN to execute its functions and may delegate selected responsibilities to either internal or external organizations in order to provide the desired end-to-end effect.
	Activities *Federated Domain Name Services Design Process (BP-454)* • Define mission name spaces • Service Design for Domain Name Service *Federated Domain Name Services Joining Process (BP-299)* • Delegate Zones *Federated Domain Name Services Change Management Process (BP-457)* • Update centrally managed zones *Federated Domain Name Services Setup Process (BP-300)* • Generate Root Key • Prepare Root Zone

Resolving Service Provider (ROLE-385)	The Resolving Service Provider operates a Domain Name Service Resolver, which is able to fully resolve a DNS resource record for a service consumer.
	Activities *Federated Domain Name Services Membership Process (BP-301)* • Provide Resolving Service *Federated Domain Name Services Joining Process (BP-299)* • Install root.hints and trusted keys
Service Consumer (ROLE-26)	An organization utilizing one or several Services. Typically, the user will require an application or the integration with own services to use the service.
	Activities *Federated Domain Name Services Change Management Process (BP-457)* • Request changes to delegated zones

Annex E. Configuration Parameters

This annex provides an overview of the Configuration Parameters (CFGs) that are necessary in order to set up the Technology Interactions (TINs) for these service instructions. The CFGs provide the granular details that must be used to setup services for testing and evaluation, training and exercises, and the whole life-cycle of participants in missions and operations. In this manner, these parameters are crucial elements for achieving day zero interoperability.

The configuration parameters are listed per TIN in a table that provides the ID, title and description of the parameter, who it is defined by (Consumer, Provider or Mission Network Service Management Authority (MN SMA)), the multiplicity (a single value or multiple entries), and optionally: a list of possible values and the default value. In many cases, the parameters that are subordinate to a broader configuration parameter are listed underneath in a bullet list.

DNS Query (TIN-26)

ID	Title	Description	Defined by	Multipl.	Values	Default value
CFG-174	root.hints	Information on the root name servers needed to initialize resolvers. The field can hold the actual information or a reference to a location where the information may be found (i.g. URL)		Single		
CFG-179	Mission Network Domain Name Space	DNS name for mission use. Mission network participants may use delegation below the Mission Network Domain Name or mission-independent names.		Single		
CFG-227	**Trust Anchors**	List of zones and trusted keys for anchoring DNSSEC validation. While there may be multiple anchors, typically this list contains only the trust anchor for the root zone.		Multiple		
CFG-229	• Trusted Key	DNSSEC key that serves as trust anchor for the given zone in DNSKEY format.	Provider	Single		
CFG-228	• Zone Name	Name of the zone for which the trust anchor applies.	Provider	Single		.
CFG-230	**List of Delegations**	The SMA has to keep a list of delegations, from which the root zone etc. are created.		Multiple		
CFG-172	• Delegated Zone	Zones to be delegated (forward and reverse).	Provider	Single		
CFG-231	• Authoritative Name Servers	List of Name Server that are authoritative for the given zone.	Provider	Multiple		
CFG-171	• Name Server	The fully qualified domain name (FQDN) of an authoritative name server for the name server (NS) entry.	Provider	Single		
CFG-170	• Name Server Addresses	The (IP) address(es) of the authoritative name server.	Provider	Multiple		
CFG-173	• Delegation Signers	Optional DS records to delegate DNSSEC signing keys to a zone. Signing keys shall be defined through both DS and DNSKEY record types	Provider	Multiple		
CFG-224	**Recursive Resolver Configuration**	Parameters and properties of the DNS Resolver(s)	Consumer	Multiple		
CFG-175	• DNS Resolver Address	IP address of the recursive resolver, which clients can use to resolve domain names.	Consumer	Single		

CFG-225	• DNSSEC Validation	Whether or not the Recursive Resolver performs DNSSEC validation.	Consumer	Single	Yes; No	No

DNS Root (TIN-370)

ID	Title	Description	Defined by	Multipl.	Values	Default value
CFG-226	**DNS Root**	Information about the DNS root, needed to bootstrap DNS resolving	MN SMA	Single		
CFG-174	• root.hints	Information on the root name servers needed to initialize resolvers. The field can hold the actual information or a reference to a location where the information may be found (i.g. URL)		Single		
CFG-227	• Trust Anchors	List of zones and trusted keys for anchoring DNSSEC validation. While there may be multiple anchors, typically this list contains only the trust anchor for the root zone.		Multiple		
CFG-229	• Trusted Key	DNSSEC key that serves as trust anchor for the given zone in DNSKEY format.	Provider	Single		
CFG-228	• Zone Name	Name of the zone for which the trust anchor applies.	Provider	Single		.
CFG-178	DNS Root Zone	Root zone for the Domain Name Services, including refresh and retry intervals, expiry and minimum TTL, top-level domains and delegations.	MN SMA	Single		

DNS Root Zone Transfer (TIN-43)

ID	Title	Description	Defined by	Multipl.	Values	Default value
CFG-220	**Zone Transfers**	Configuration of a DNS Zone Transfer	Provider	Multiple		
CFG-221	• Zone Name	Full name of the zone to be transfered	Provider	Single		
CFG-176	• Transfer Source	Address of the authoritative name server hosting the original zone file.	Provider	Single		
CFG-177	• Transfer Destination	Address of the authoritative name server that receives the zone file.	Consumer	Single		
CFG-222	• Key name	TSIG NAME record - i.e. a mnemonic key name in domain name format	Provider	Single		
CFG-223	• Shared Secret	The key itself, which is used to authenticate the zone transfer	Provider	Single		

Anycast DNS Advertising (TIN-114)

ID	Title	Description	Defined by	Multipl.	Values	Default value
CFG-633	**Anycast DNS Service**	The anycast service configuration holds all all required parameters to configure Anycast DNS.		Multiple		

CFG-635	• Anycast ASN	The Anycast Autonomous System Number is the agreed ASN to be used for the Anycast DNS service.		Single		
CFG-634	• IP Address	The IP address(es) to be used for the Anycast DNS Service	N/A	Multiple		